```
$ npm run dev -w ch07/summarization-web-app
```

You'll also need either the OpenAI API or the Google API keys to access their REST API. For details on setting up API keys, refer to the appendix.

```
// .env
OPENAI_API_KEY=<OPEN_AI_SECRET_KEY>
GEMINI_API_KEY=<GOOGLE_AI_SECRET_KEY>
```

NOTE The core LangChain.js library is implemented in TypeScript/JavaScript and does not require Python to be installed on your system. It runs natively in Node.js or browser environments. However, some community integrations or example projects may depend on Python tools or libraries and thus require a Python environment.

The following figures are screenshots of the new UI when running the application. In the first use case (figure 7.4), the user uploads either a PDF or a DOCX document, and the AI provides a summary of this document.

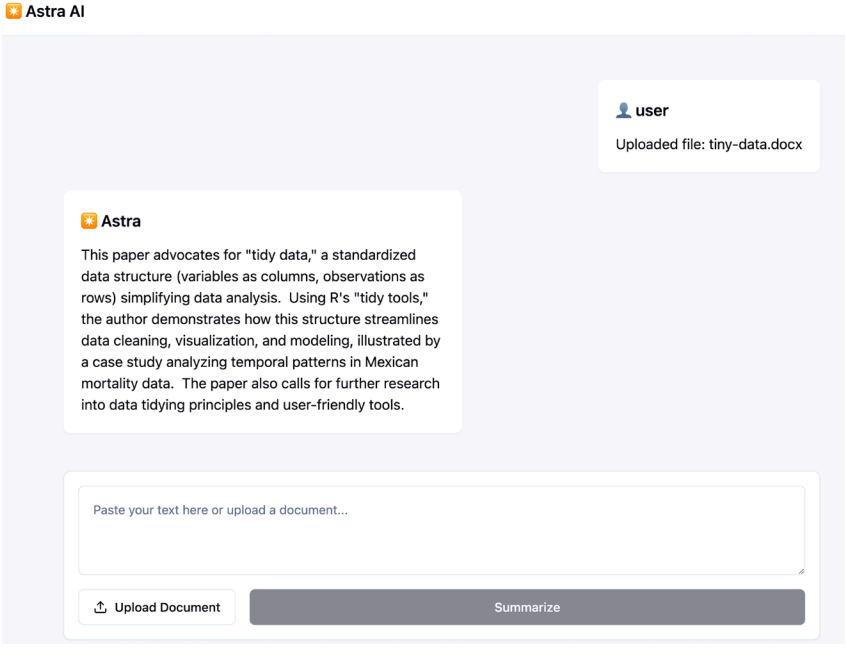


Figure 7.4 How the web application uses a MapReduce summarization technique to generate a concise summary from a large document through the UI. The application then sends the input to the backend service, which uses the LangChain.js library to split the document into manageable chunks, process them through the summarization model, and return the final summary to be displayed in the user interface.